THE BIOLOGY OF AGING AND AGE-RELATED DISEASES

NEW DIAGNOSTIC, PREVENTATIVE AND HEALING STRATEGIES FOR AGE-RELATED DISEASES

(Proposed by Miroslav Radman
To be further elaborated and completed)

The Basics:

Life is a complex cellular structure exerting a complex orchestrated function fueled by cell-made energy. Protein molecules carry out all life's functions whereas DNA is life's information to make proteins (by dedicated proteins and their RNA products!).
Proteins do everything in life (including maintenance and "reading" of the genes) by synthesizing large and small molecules that do, or complete, what proteins don't do directly. Genes dictate which and how much of each of the thousands of proteins is being made in each cell. Hence, proteins and genes are the "chicken and egg" of life.

This is the context to see any disease. Disease is a defect in life's biochemical harmony (or homeostasis). Since proteins do directly or indirectly all biological work, a disease as biological dysfunction must result from a defect in protein function as its root cause. Such defect can reside in genes - information to make proteins - arising by mistakes in copying genes (mutations) or in copying their chemical "decoration" (epigenetic changes that determine how active the genes are). Both kinds of gene alterations occur because relevant proteins made mistakes (ref 1). When genes carry mutation that ruins protein function, then the baby is born sick (or is not born alive) and we call such diseases the syndromes. Dysfunction can arise and develop to various degrees also by accumulation of direct damage to the fine structure of proteins with its functional consequences. Aging appears as a snowballing of biological dysfunctions. Even dysfunctions can snowball harmoniously (healthy aging) whereas diverse age-related diseases are disharmoniously snowballing dysfunctions.

Proteins are fragile (average protein lifetime in human cells is about ten hours) and must be constantly replenished starting from RNA copies of their genes. Thus, even with intact genes, mistakes in protein production (synthesis and folding) and direct chemical damage to proteins can ruin their function (to varying extent) and cause progressive aging and age-related disease.
Thus, damage to relevant protein has repercussions upon the quality of genes (mutations and epigenetic changes). Chicken-egg effect. This is the simple conceptual context in which, at the outset, any disease can be considered. The rest are details that we need to know about for medical interventions.

The increase with age in the incidence rate of all age-related diseases and the existence of healthy centenarians argue for a likelihood of a common cause of aging and all age-related diseases - probably the accumulating protein damage. By preventing and decreasing such

common cause, it should be possible to mitigate all age-related diseases simultaneously and slow-down the rate of aging. Then, most people could live the life of fellow supercentenarians - up to 120 years - *without changing the basic human biology* (until we acquire relevant knowledge to extend the lifespan beyond 120 years by changing human biology)!
Below, we tackle the problem of dealing with inborn predisposition to age-related diseases.

New Principles and Approaches to Disease Diagnostics and Treatments

The independent, private, not-for-profit Mediterranean Institute for Life Sciences (*MedILS*) in Split, Croatia, and its founder Miroslav Radman, have pioneered the research on molecular biology of aging in the continental Europe and developed a new approach in the study of aging and age-related diseases. The results of over a decade of our research are:
(a) the discovery of the common root cause of aging and age-related diseases (ref 2),
(b) a principle for the diagnostic identification of inborn predispositions to age-related diseases (cancer, cardiovascular, neurodegenerative and immunity-related diseases) with the concept for mitigation of such diseases by attenuating the pathological consequences of genetic predispositions (the way they are attenuated in youth, ref 1,2)
(c) uncovering of a cellular tissue-level mechanism of the long latency period before the onset of age-related disease (called *cellular parabiosis*) and how disease latency could be further extended (ref 3).

The root cause, or “the chemistry”, of aging is oxidative damage to the proteins (protein corrosion) and the inborn predispositions to diseases are “silent mutations” in genes (amino acid substitutions in proteins), i.e., genetic “polymorphisms”, shown to sensitize the disease-relevant proteins to malfunction via their increased oxidation (ref 2). We have shown that any kind of protein folding or biosynthetic errors in proteins increase their intrinsic sensitivity to oxidation perturbing their biological function (ref 1,2). Hence, only the native, perfectly folded proteins display the evolved resistance to oxidation.

Protein oxidation is reversible and so are its pathological consequences (the principle of the Project A). Few irreversible consequences of oxidatively damaged proteins are mutations and epigenetic alterations. However, pathological consequences of such irreversible changes in DNA are subject to “phenotypic suppression”, i.e., functional compensation by the surrounding cells called *cellular parabiosis* (ref 3).

Today, such diseases account for 90% of morbidity and mortality in Europe and USA and progressively so on other continents. Common root cause means that it should be simpler and easier to prevent simultaneously all age-related diseases together than any particular one. The robust health of aging centenarians and super-centenarians proves this point: usually, at 90 years of age, although slowed down in their mobility, they live active and intellectually productive life and do not take medications.

THE PROJECTS:

(A) Predictive diagnostic of the innate predisposition to age-related diseases and follow-up (monitoring) of relevant biomarkers (collaboration of Drs. Guillaume Combes, Katarina Trajkovic and Mladen Mercep and their respective teams.)
Reference 2 and ongoing research show that the fragility (oxidability) of specific proteins can be used to identify the cause of predisposition to Parkinson's, ALS and probably other age-related diseases. This is a breakthrough that will be developed into a routine disease diagnostic and follow-up biomarker for monitoring the state of inborn "weak links". Neurodegenerative and malignant diseases and diabetes will be given the priority.

(B) Prevention of the cause of aging and age-related diseases and treatment of manifested diseases: search for chemical antioxidant chaperones that protect fragile proteins from oxidative damage. (Dr. François-Xavier Pellay and his group)
We have found first natural compound protector against oxidative damage of the most sensitive proteins which was isolated from an extremely radiation-resistant microorganism. The protector molecule is a non-toxic anti-oxidant chemical chaperone of proteins. We propose to look for other similar molecules in extremely robust microbial, plant and animal (ref 4) species and have set up a screening system for such molecules (Dr. François-Xavier Pellay).

(C) Keeping age-related diseases in latency: search for molecules that protect or stimulate cellular parabiosis (the team of Dr. Katarina Trajkovic)
Cellular parabiosis is a trans-cellular functional complementation of cellular defects by the molecular traffic between the cells in tissues (tissue homeostasis via tunneling nanotubes, gap junctions, cargo vesicles, etc.). Cellular parabiosis can prevent phenotypic expression of deleterious DNA alterations and fully reverse already expressed cellular defect (ref 3). Chronic Inflammation prevents such "cellular solidarity". Anti-inflammatory molecules that preserve, or stimulate, cellular parabiosis can be screened by a system set-up by my colleague Dr. F-X Pellay.

(D) Specificity and tissue selectivity of cancer metastases (Dr. Andrea Golemanovic and graduate student Tinka Vidovic).
Cellular parabiosis paradigm provides an obvious hypothesis that the same mechanism of latency (20-25 years in humans) in phenotypic expression (onset) of initiated solid primary cancers (carcinomas) prevents growth of their disseminated cells to establish metastases. First approach uses bioinformatics to explore whether the metastases can readily grow only in tissues that do not express "driver" tumor suppressor genes that are inactivated by mutation in the primary tumor. In such tissues, no phenotypic suppression of malignant growth via cellular parabiosis is expected. If the results encourage the hypothesis, an experimental proof of principle will be undertaken with tumor cells bearing identified "driver" tumor suppressor mutations and isogenic fibroblasts with the same either active or inactive tumor suppressor genes as "feeder layers" for growth of individual tumor cells (ref. 5).

(E) Visualization and detection of all newly emerging mutations in individual human cells (graduate student Vlatka Matkovic supervised by Dr. K. Trajkovic)
We have already developed a unique fluorescent microscopy method for visualizing all individual genome-wide mutations (in single cells and in real time) that emerge during DNA replication in *E. coli* bacteria (ref 6). The method uses a functional fluorescent mismatch repair protein MutL (ref 6). Now, we use human MutL homolog protein hMLH1 and monoclonal antibodies (for immunofluorescence) to see mutations in human cells. We can observe signals for 18 (+/- 4) mutations per replication of the human diploid genome, i.e., at each genome replication 18 new mutations are added. After appropriate controls, we intend to isolate millions of small DNA fragments bearing single new mutations to be identified by sequencing and mapped on the genome. In other words, we wish to establish the place in genome and the nature of the ensemble of emerging human mutations (human "mutome").